

Gao Wei

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◇ Address: University of Macau, Taipa, Macau SAR, China

EDUCATION

Wuhan University	Bachelor of Engineering
Majoring in Electronic Information Engineering	09/2016 - 06/2020
University of Macau	Master of Science
Majoring in Artificial Intelligence applications	08/2021 - 05/2023
University of Macau	Ph.D.
Majoring in Computer and Information Science	08/2023 -

RESEARCH EXPERIENCE

Intelligent Robotic Nursing Assistant	Imperial College London, The Hamlyn Centre
Summer School	07/2019 - 08/2019

- Intelligent nursing system composed of voice module, 3D camera, and robot arm.
- The main technologies involved are speech recognition, face recognition, target recognition, and robotic arm motion trajectory algorithm.
- The project finally realizes the function of automatic feeding and picking up objects by the robotic arm.

Future Urban Intelligent Transportation Augmented Emulation Platform	University of Macau, IOTSC
	08/2021 - 05/2023

- Development of systems including VR simulated driving systems, digital twin systems, SLAM-based autonomous driving systems, and autonomous tracking driving systems

A framework of Robot Exploratory SLAM for 3D Mapping	University of Macau, IOTSC
	05/2023 - 08/2024

- A real-time framework for autonomous robotic mapping in large-scale urban scenes with OpenStreetMap (OSM) as guidance.

3D AgentWorld Lab	The Hong Kong University of Science and Technology (Guangzhou)
Research Assistant	08/2025 - 12/2025

- Contributed to three research manuscripts on simulation scene generation, embodied navigation, and autonomous decision-making for robot learning (1) Scalable Simulation Scene Generation from a Single Image via a Multi-Agent Framework for Robot Learning (2) Causal-Driven Active Uncertainty Mitigation for Zero-Shot Visual Navigation (3) Agentic Zero-Shot Navigation via Bayesian Context and Active Verification

PUBLICATIONS

- Wei Gao, Zezhou Sun, Mingle Zhao, Chengzhong Xu, and Hui Kong, Active Loop Closure for OSM-guided Robotic Mapping in Large-Scale Urban Environment, IEEE International Conference on Intelligent Robotics and Systems (IROS), 2024. Project GitHub. Project video
- Yuming Huang, Wei Gao, Zhiyuan Zhang, Maani Ghaffari, Dezhen Song, Chengzhong Xu, and Hui Kong, Learning Orientation Field for OSM-Guided Autonomous Navigation, Journal of Field Robotics (JFR), 2025 Project GitHub Project video
- Wei Gao, Jie Zhang, Mingle Zhao, Zhiyuan Zhang, Shu Kong, Maani Ghaffari, Weisong Wen, Dezhen Song, Chengzhong Xu, and Hui Kong, Super LiDAR Reflectance for Robotic Perception, IEEE Robotics and Automation Letters (RAL), 2026 Project GitHub Project video
- Wei Gao, Jian Shu, Mingle Zhao, Maani Ghaffari, David Kong, Chengzhong Xu, and Hui Kong, Cyclops: LiDAR as a Camera That Dreams in Color, The Conference on Robot Learning (CoRL), under review, 2026

PATENTS

- An autonomous driving test platform based on virtual-reality integration technology. CN Patent CN117191419A. Filed August 25, 2023, and issued December 08, 2023.